Abstract

The present invention discloses a three-layer separated multi-agent system operation system and method, comprising a **Rule Cognition Layer**, an **Isolated Asset Layer** and a **Computing Execution Layer**.

The **Rule Cognition Layer** is physically deployed in a **trusted isolated operating environment**. It allocates a temporary **isolated secure operating domain** and starts a hardware lifetime timer **Timer_TTL**. Private knowledge slice data is read from the locally deployed **Isolated Asset Layer** via a **unidirectional read-only physical tunnel**. Instruction assembly is completed inside the **isolated secure operating domain**, and a causal **instruction anti-replay digital fingerprint** is generated by coupling microsecond timestamp **T_μs**, which is unidirectionally output to the **Computing Execution Layer** deployed on the cloud inference platform. When the timer expires or physical boundary penetration is detected, the chip-level security controller forcibly triggers a **physical degaussing erasure** interrupt, performs multiple rounds of memory overwriting **N_scrub**, and invokes the clflush instruction to clear the processor cache and complete charge neutralization.

The **Computing Execution Layer** monitors data throughput during inference. When an **Over-ingestion Attack** (the technical implementation of Semantic Binge at the system level) occurs and traffic exceeds the buffer upper limit **Theta_max**, the session is terminated, and a TCP RST packet carrying the dedicated network forensic fingerprint **W_forensic** is forcibly transmitted externally.

The present invention realizes complete decoupling of knowledge base data, logic control and inference computing power at both physical and network levels. It can effectively block side-channel cold boot data extraction and unauthorized **Over-ingestion Attacks** targeting cloud multi-agent nodes, and safeguard the data sovereignty of users' private knowledge.